



Genesis Biotech Inc.

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TNF (Tumor necrosis factor)-alpha

Catalog No. PG-40005

Scale: 100 µg

Protein species: Human

Specific activity: 1×10^7 IU/mg

Applications tested: cell culture

Expression Host: *E. coli*

Type: purified recombinant protein

Form: sterile filtered white lyophilized (freeze-dry) powder

Target description

Tumor necrosis factor (TNF) family includes two structurally and functionally related proteins, TNF- α or cachectin and TNF- β or lymphotoxin. TNF- α is produced mainly by monocytes and/or macrophages, whereas TNF- β is a product of lymphoid cells. Originally thought of as selective anti-tumor agents, TNF- α or TNF- β (especially the former) are now grouped among the major "inflammatory cytokines," they are characteristically produced at the sites of inflammation by infiltration mononuclear cells. They play a beneficial role as immunostimulants and important mediators of host resistance to many infectious agents and, probably, malignant tumors.

Potential applications

1. An antineoplastic agent for the treatment of patients with locally advanced solid tumor.
2. It is a potent pyrogen, causing fever by direct action or by stimulation of interleukin-1 secretion.
3. It can stimulate cell proliferation and induce cell differentiation under certain conditions.
4. It can cause cytolysis of tumor cell lines.

Comments

TNF(Tumor necrosis factor): Molecular Mass: 17.5kDa, composed of 158 aa. Human TNF is recombinant DNA produced and >95% pure via SDS-page and HPLC.

Related Products

1. IGF-1(Insulin-like growth factor) (PG-40001)
2. PDGF-B (platelet-derived growth factor B-chain) (PG-40002).
3. bFGF (basic fibroblast growth factor) (PG-40003)
4. Fibronectin (PG-40004)

Antibiotic residue

No activity of ampicillin or other antibiotic residue be detected.

Bacterial endotoxin content

Bacterial endotoxin content is not higher than 0.1ng/ug (1EU/ug)

Specific Activity

Specific activity is defined as the activity divided by the mass of the protein sample. Specific activity: 1×10^7 IU/mg

Solubility

It is recommended to reconstitute the lyophilized TNF- α in sterile H₂O not less than 100µg/ml, which can then be further diluted to other aqueous solutions.

Storage

For short-term use store at 4 °C. For long-term storage keep at -20°C.

References

1. Gallily R, Sher T, Ben-Av P, Loewenstein J. Tumor necrosis factor as a mediator of Mycoplasma orale-induced tumor cell lysis by macrophages. Cell Immunol. 121(1):146-53, 1989.
2. Lung HL, Shan SW, Tsang D, Leung KN. Tumor necrosis factor-alpha mediates the proliferation of rat C6 glioma cells via beta-adrenergic receptors. J Neuroimmunol. 166(1-2):102-12, 2005.
3. Conti B, Tabarean I, Andrei C, Bartfai T. Cytokines and fever. Front Biosci. 1;9:1433-49, 2004.

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